



Harvesting Insights: Usability Study of the USDA ACIR Portal

Final Report

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SI 622 - Needs Assessment & Usability Evaluation

ACIR is the public-facing entry point for U.S. agricultural import rules.

- Over 100,000 country and commodity combinations
- About 16,000 Commodity Import Requirement documents
- Built on a Salesforce platform

An official website of the United States government [Here's how you know](#) >

USDA Animal and Plant Health Inspection Services
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Agricultural Commodity Import Requirements

APHIS Import Requirements Search Tool for Plant, and Agriculturally Related Products

ACIR Newsletter

The ACIR Team publishes a quarterly ACIR Newsletter that keeps users informed about system enhancements, new tools or tips for using the portal, and other ACIR related news.

Check out the newest issue published 4/13/2026 - FY26 Q2 ACIR Newsletter

< To sign up to get this publication emailed to you:
Go to the [USDA APHIS Stakeholder Registry](#) and select "Agricultural Commodity Import Requirements (ACIR)" under Plant Health Information, News and Information, Electronic Databases - System Status.

To check past issues, visit the ACIR Newsletter Archive on the [Help](#) page in the About ACIR section (2.4).

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Plants for Planting and Propagation

Imported material of plants and vegetative parts that are for or capable of propagation, including buds, corns, cuttings, layers, pollen, scions, seeds, tissue, tubers, and like structures



Plants and Plant Products Not for Propagation

Fruits, vegetables, seeds, cut flowers, or handicrafts not for propagation, including products that are fresh, cooked, dried, cut, frozen, and other similar processed states



Wood Products and Byproducts

Logs, lumber, cross ties, firewood, wood chips, mulch, wood packaging material, cork and other similar articles



Pest Organism Permitting

Fungi, insects (including butterflies, bees, and moths), regulated organisms used in pet trade (or smuggled), annelids (earthworms), parasitic plants, bacteria, noxious weeds and more. See also nonregulated organisms.



Other Items

Tools, materials, or machinery that can act as carriers for pests, including vehicles, containers, bags, packing materials, and other similar articles



Soil, Soil Constituents and Rocks

Compost, manure, rock, sand, soil, and other similar articles



Treatments

Phytosanitary treatments for quarantine-significant plant pests on imported commodities

Primary users



Example use-case

Maria, Customs Broker

She handles many commodity types each week. Plants for planting are a small part of her work. She needs an answer in under five minutes.

Scenario: A client asks Maria whether a specific seed from Italy can enter the United States. She opens ACIR, types the scientific name, and the result is not what she expected.

Research Methods

1. 6 semi-structured interviews
2. Survey instrument designed and piloted
3. Comparative evaluation against
eCFR, APHIS Manuals, Microsoft Learn, and GOV.UK
4. Heuristic evaluation with 5 evaluators using Nielsen heuristics
5. Usability test with 1 pilot and 7 moderated sessions

The findings shown next appeared in at least three of these methods.

Finding 1: Search fails on imperfect input.

Users fall out of the workflow when terms do not match exactly.

This pattern appeared in every method. Heuristic evaluators rated it severity 4 (5 of 5). Interviewees reported autocomplete dropping out on full names and no way to search by country alone. Comparators (eCFR, Microsoft Learn, GOV.UK) all support free text and spelling correction. In testing, a 20 year veteran hit friction on Task 3 when a scientific name returned no match, and a super user left Task 3 incomplete due to undefined terms.

Recommendation: Make search forgive small mistakes.

Add forgiving autocomplete that accepts partial input, common names, and synonyms. Surface taxonomy suggestions inline when a query has no exact match. Support country only and commodity only queries. These changes lower the cost for new users without slowing experts down.

Finding 2: Help is disconnected from the task.

Users must leave their workflow to find context, definitions, or guidance.

Help content lives across the Help page, Videos and Guides Hub, glossary, manuals, and homepage banner with no single entry point. Heuristic evaluators rated this severity 4 (5 of 5). The comparative evaluation showed eCFR embeds definitions inline and Microsoft Learn uses related links sidebars. In usability testing, even experienced users could not reliably connect manual guidance to the matching CIR document. One participant called Task 2 "taxing" due to disconnected documents, and another flagged undefined terms like PCN country with no inline explanation.

Recommendation: Bring help into the task.

Consolidate manuals, videos, and glossary into a single Help Hub. Add hover tooltips for technical terms inside CIR documents and search forms. Add direct links between manual sections and matching CIR documents so users do not rely on prior knowledge to connect them.

Finding 3: Small inconsistencies misplace user trust

Conflicting tile data, lost search state, and blank transitions lower confidence in the answer.

Interviewee 5 described a case where *Angelonia angustifolia* was classified differently across tiles, leading an officer to give incorrect guidance and call back to reverse it. Heuristic evaluators flagged missing breadcrumbs (severity 3), no loading feedback (severity 3), and inconsistent link behavior. Interviews also surfaced inconsistent regulatory language across entries, where some commodities had clear prohibitions and others used ambiguous permit language. These small breaks accumulate, and users stop trusting what they find.

Recommendation: Close the trust gaps.

Add cross reference checks that flag conflicting tile classifications before they reach users. Standardize status language across CIR entries so admissibility is unambiguous. Add breadcrumbs, consistent link behavior, and loading indicators so users always know where they are and what the system is doing.

Discussion

Areas this study could not fully cover

- Internal officer-only views that require logging into the portal
- Mobile and small form factor experience and testing
- Accessibility audit against WCAG standards
- General public users such as travelers and home gardeners

Suggested next research steps

- Recruit interview and usability test participants for a follow-up usability test
- Run a card-sorting study to redesign the help architecture
- Prototype a more forgiving search and test it with users
- Deploy the survey at scale to confirm pain points across a wider sample

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Across interviews, a piloted survey, a comparative evaluation, a heuristic evaluation, and seven moderated usability tests, three patterns held across every method. ACIR is a thorough and authoritative source of import requirements, and the people who use it most rely on it daily. The portal struggles in three specific places. It is hard to enter the system without insider knowledge. It is hard to get help in the middle of a task. And small inconsistencies misplace trust in the answer.

Make the portal forgive small mistakes.

Every method showed users falling out of search the moment their input did not match an exact scientific or approved name. Forgiving autocomplete, partial matches, and synonym surfacing would lower the cost of being a new or occasional user without slowing experts down. Plain-language sub-labels under the homepage tiles would help users pick the right path the first time. These changes meet new users where they are while keeping the precision experienced users depend on.

Bring help into the task.

Help content lives in many places, and our usability test confirmed that even experienced users cannot reliably connect manual guidance to the right CIR document. A consolidated Help Hub, hover tooltips for technical terms, and direct links between manual sections and matching CIR documents would put context where users actually need it. The goal is for users to never have to leave their task to understand it.

Close the trust gaps.

Conflicting tile classifications, blank loading transitions, and lost search state after the browser back button all lower confidence. Cross-reference checks for tile data, breadcrumbs, standardized link behavior, and clear loading and confirmation messages would promote credibility.

The evidence (reports, presentations, and videos) for every assignment is documented in our final report which was compressed and sent as a ZIP file to the USDA APHIS team.

Appendix

- [Link to Video Presentation \(YouTube\)](#)

Summaries for Interview, Comparative Evaluation, Heuristic Evaluation, and Usability Testing in next slides.

Interviews

Research Goal

Understand how people find, interpret, and act on import requirements in their daily work, and uncover the pain points and workarounds that shape their experience.

Interview Design

6 semi-structured interviews, each 45 to 60 minutes. Twelve open-ended questions focused on search workflow, comprehension, and collaboration. Sessions were recorded, transcribed, and coded by at least two team members.

Participants

Policy managers, PPQ officers, a training specialist, and one importer. Recruited directly by USDA and representing multiple departments across different states.

Key Findings

Five themes stood out. The PDF to hyperlink transition hurt scannability. Regulatory language is inconsistent across entries. Contradictory tile data led to incorrect guidance in the field. Autocomplete drops out on full-name input. High-volume importers bypass ACIR for downstream systems like e-file.

Recommendations

Offer a scan-optimized view next to the compliant version. Standardize how requirement language is presented. Flag conflicting classifications across tiles. Fix autocomplete behavior for full-name input. Explore tighter integration between ACIR and e-file.

Comparative Evaluation

Research Goal

Benchmark ACIR against similar products to learn how other systems help users navigate dense regulatory or technical information.

Comparators

Four products across three comparator types. eCFR as a direct government comparator, APHIS Manuals as a partial comparator, Microsoft Learn and GOV.UK as analogous comparators with strong information architecture.

Evaluation Criteria

A comparative matrix covering navigation, search flexibility, information hierarchy, scannability, terminology clarity, accessibility, and interaction style.

Key Findings

Four gaps: ACIR search is rigid compared to competitors that support free-text and partial queries. Related content is not linked inside results, while eCFR, Microsoft Learn, and GOV.UK all cross-link. Terminology assumes expert knowledge with no inline definitions. Scannability lags behind GOV.UK and Microsoft Learn, which use clear hierarchy and action-first layouts.

Recommendations

Support free-text and partial queries with filters. Add related content panels and inline links to treatments, facilities, and taxonomy terms. Add plain-language tooltips or a built-in glossary like eCFR's definitions section. Improve page hierarchy for faster scanning.

Heuristic Evaluation

Research Goal

Identify and rank design-level usability violations in the public-facing portal using expert inspection.

Method

All five team members independently evaluated ACIR against Jakob Nielsen's ten usability heuristics. Before starting, we held a calibration session on a shared set of screens. Severity ratings used a zero to four scale from cosmetic to catastrophic.

Process

Evaluators documented each issue with its violated heuristic, location, and severity. After independent review, we aggregated findings, merged duplicates, and assigned consensus severity through a group debrief.

Key Findings

Sixty-plus issues were narrowed to five findings, each flagged by at least four of five evaluators. Search rigidity, severity four. Scattered help content, severity four. Navigation disorientation with no breadcrumbs, severity three. No loading feedback, severity three. Regulatory jargon and inconsistent tile labels, severity three.

Recommendations

Move to a faceted search with save-search and persistent autocomplete. Consolidate manuals, videos, and glossary into one Help Hub with inline tooltips. Add breadcrumbs and standard link behavior. Add loading indicators and confirmation messages. Audit tile classifications and add plain-language sub-labels.

Usability Testing

Research Goal

See whether real users can complete realistic import lookup tasks on ACIR, and identify where they get stuck.

Test Design

Moderated, task-based, think-aloud testing. Seven tests, each about 45 minutes, hosted via Zoom. Each session was moderated by one facilitator and one notetaker.

Participants and Tasks

Six regulatory professionals with varying ACIR experience. Three tasks: Determine if an apricot from Mexico is admissible. Find small lots of seed guidance for *Solanum lycopersicum* from Italy and connect it to the matching CIR document. Verify a scientific name and find admissibility and quarantine info for *Leucanthemum nipponicum* from Dominica.

Key Findings

Three patterns held across sessions. Small input mismatches and overlapping result sets caused friction even for experienced users. The link between manual guidance and the CIR document was rarely obvious, leading to long detours through PDFs. The taxonomy page was hard to find when a scientific name did not match the search field.

Recommendations

Build forgiving autocomplete that accepts common names and synonyms. Surface a direct link to the matching CIR document from each manual section. Surface taxonomy and synonym data inside the search field so users do not have to open a separate page.